COMPUTER SCIENCE
THESIS 1

SHORT MODULE

HOW TO CHOOSE A THESIS TOPIC IN COMPUTER SCIENCE

A practical guide from broad interest to
a focused, researchable, and feasible topic.

1. What Makes a Good Computer Science Thesis Topic?

- It begins with a clear problem - not merely with a technology you want to use.
- It contains a computational task that can be investigated, tested, or improved.
- It uses data, evidence, experiments, or measurable observations.
- It has a manageable scope that can be completed within your available time and resources.
- It produces a contribution that can be explained and defended.

Remember:

A thesis investigates. A project mainly builds.

2. Start With a Problem, Not With a Technology

WEAK START

> **"I want to use Artificial Intelligence."**

BETTER START

> **What problem needs to be predicted, detected, classified, optimized, recognized, or recommended**

Example progression:

Interest:	Artificial Intelligence
Domain:	Education
Problem:	Students at risk of failing programming
CS Task:	Early prediction
Study:	Compare machine-learning models

3. Narrow the Topic: Broad to Researchable

TOO BROAD

Artificial Intelligence

BETTER

AI for Education

FOCUSED

Student Performance Prediction

RESEARCHABLE

Comparative evaluation of machine-learning models for early prediction of at-risk students

4. Find the Computational Core

Ask: What exactly should the computer do better?

- **Classification** Assign something to a category
- **Prediction** Estimate an outcome or value
- **Detection** Find an object, event, or anomaly
- **Optimization** Find the best solution among alternatives
- **Recommendation** Suggest the most relevant option
- **Clustering** Discover groups or patterns in data
- **Information Retrieval** Find the most relevant information

5. Check Data and Feasibility Before Finalizing

DATA

Where will the data come from?
Public dataset? Images? Logs? Sensors? Institutional records?

ACCESS

Can you legally and ethically obtain the data?
Do you need permission or ethics review?

RESOURCES

Do you have enough time, hardware, software, budget, and technical skill?

SCOPE

Can the study be completed within one thesis period?

6. Identify the Research Contribution and Evaluation

Possible contributions

- Improve accuracy, speed, robustness, or efficiency
- Compare algorithms under a specific context
- Combine methods in a meaningful way
- Adapt a method to a constrained environment
- Develop or curate a useful dataset

Possible evaluation measures

- Classification: Accuracy, Precision, Recall, F1-score
- Regression / Prediction: MAE, RMSE, R-squared
- Algorithms: execution time, memory use, solution quality
- HCI: task time, error rate, usability, workload

Your conclusion should be stronger than:
"The system works."

7. Project Idea vs. Thesis Topic

PROJECT IDEA

Parking Management System

STRONGER RESEARCH DIRECTION

Computer vision-based parking slot occupancy detection using lightweight object detection models

PROJECT IDEA

Job Portal

STRONGER RESEARCH DIRECTION

Semantic job recommendation using sentence embeddings and skill similarity analysis

PROJECT IDEA

Crop Monitoring System

STRONGER RESEARCH DIRECTION

Evaluation of lightweight deep-learning models for mobile crop disease classification

8. Use the F.I.T. Test

F

FEASIBLE

Can you finish it using the time, data, people, tools, budget, and computing resources available to you?

I

IMPORTANT & INTERESTING

Is the problem worth investigating? Will the result be useful or add knowledge?

T

TESTABLE

Can you evaluate the proposed method and produce measurable evidence?

A narrow problem studied well is better than a huge problem studied superficially.

9. Build Your Working Thesis Topic

1. **Problem**
   What real problem will you investigate?

2. **Data**
   What evidence or dataset will you use?

3. **Method**
   What computational approach will you study?

4. **Evaluation**
   How will you measure performance?

Simple formula

PROBLEM + DATA + METHOD + EVALUATION
= A POTENTIAL COMPUTER SCIENCE THESIS

10. Final Topic Checklist

- [ ] I can state the specific problem in one or two sentences.
- [ ] The study has a clear Computer Science or computational component.
- [ ] I know what data I need and how I can obtain it.
- [ ] I can explain what will be compared, improved, tested, or evaluated.
- [ ] I know the metrics or evidence I will use.
- [ ] The scope is realistic for the available time and resources.
- [ ] The contribution is more than simply developing an application.

Short Activity: From Interest to Thesis Topic

Prepare THREE possible thesis ideas. For each idea, write:

- **Area of Interest**
- **Real-World Problem**
- **Target Users / Environment**
- **Available Data**
- **Computational Task**
- **Possible Method**
- **Evaluation Metrics**
- **Possible Research Contribution**
- **Working Thesis Title**

Then choose your strongest idea and answer:

Why is this problem worthy of a Computer Science thesis?

Write a short 3-5 sentence justification.